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Project part 1

Q1

What do you foresee for the future growth of the Renewable Energy industry? What impact do you believe this might have on the situation with data at Helios Energy?

In my opinion, the renewable energy industry is not just growing. It is accelerating really fast and becoming the core of future energy infrastructure. Governments, investors, and companies are all pushing heavily toward solar, wind, and battery storage.

According to the **International Energy Agency (IEA)**, global renewable capacity is expected to grow by nearly 2.5 times between 2022 and 2030, with solar PV leading most of that growth.[1] Also, the **U.S. Energy Information Administration (EIA)** projects that renewable energy will be the fastest-growing source of electricity generation in the United States through 2050.[2]

On top of that, global investment in clean energy surpassed $1.7 trillion in 2023 according to the IEA. [3] That shows this is not temporary hype, it is long-term structural growth.

Impact on Data at Helios Energy

Now relating this to Helios Energy:

If the renewable industry grows this fast, Helios will likely:

- Expand into new markets
- Launch more product variations (solar panels, inverters, storage units)
- Onboard more customers and partners
- Generate much more operational and IoT data

Right now, based on the case, Helios already struggles with Product Master inconsistencies and customer data issues. If growth continues without fixing data governance, the problems will multiply.

For example:

- More products = more duplicate SKUs
- More installations = more customer data duplication
- More regions = inconsistent naming conventions
- More regulatory pressure = higher compliance risk

In my opinion, fast growth without strong data architecture will create operational chaos. Instead of benefiting from expansion, Helios could face:

- Reporting inaccuracies
- Inventory errors
- Poor forecasting
- Delayed decision-making

So basically, renewable growth increases opportunity but it also increases data complexity exponentially. If Helios does not fix its data foundation now, scaling will make everything worse.

Footnotes

1. International Energy Agency (IEA), *Renewables 2023 Report*, 2023.
2. U.S. Energy Information Administration (EIA), *Annual Energy Outlook 2023*, 2023.
3. International Energy Agency (IEA), *World Energy Investment 2023*, 2023.

Q2

What would you recommend Helios Energy do to address the limitations related to the Product Master? What are the benefits and risks?

In my opinion, Helios needs to treat Product Master data as a strategic asset, not just an operational file in the ERP.

My Recommendation

I recommend three main actions:

1. Implement Master Data Management (MDM)

Helios should centralize the Product Master into a single controlled system (MDM platform), instead of allowing multiple departments to maintain separate versions.

That means:

- One golden record per product
- Standardized SKU structure
- Standard naming conventions
- Clear ownership of product data

2. Establish Data Governance Policies

They need:

- Defined data stewards
- Approval workflows before new product creation
- Version control
- Mandatory validation rules (no incomplete attributes allowed)

Right now it seems like anyone can create or modify product entries, which causes duplicates and inconsistencies.

3. Create Product Lifecycle Controls

From product creation → updates → retirement, there must be structured workflow. No more ad-hoc updates.

Benefits of My Recommendation

In my opinion, the benefits are very clear:

- Better inventory accuracy
- Improved forecasting and demand planning
- Reliable financial reporting
- Better analytics for growth decisions
- Reduced duplication and operational waste

Also, as Helios scales in the renewable industry, clean product data allows faster integration with:

- Suppliers
- Installers
- CRM systems
- IoT systems

Risks of My Recommendation

But it’s not risk-free.

1. Implementation Cost

MDM systems and governance processes require investment.

2. Resistance to Change

Employees may resist stricter controls.

3. Short-Term Disruption

During data cleaning and migration, operations could slow temporarily.

4. Over-Engineering

If governance becomes too rigid, it might slow innovation.

My Overall View

In my opinion, not fixing Product Master is riskier than fixing it.

Renewable energy growth will increase product complexity. If Helios doesn’t standardize now, their data issues will compound and hurt profitability, reporting accuracy, and scalability.

I’d rather take short-term disruption and fix the foundation than scale on messy data.